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16.122 Aerothermodynamics

DESIGN PROJECT

Issued: April 17, 2026

Due: May 5, 2026

Focus Areas: Inviscid fluid, viscous fluid, shock waves, pressure distribution, heat transfer, material selection, boundary layers, vehicle size and weight

Objective: Design a hypersonic commercial aircraft connecting Uppsala, Sweden and Singapore. *Depart Uppsala, fly over Gulf of Bothnia, Greenland, Arctic Ocean, Pacific Ocean, South China Sea to land in Singapore*. Do NOT consider takeoff and decent profiles.

Configuration Modeling: (a) composite of flat plates, wedges, and sharp cones or
(b) curved surfaces or
(c) combination of (a) and (b) or
(d) other

Propulsion Modeling: (a) hydrogen fuel or
(b) turbojet or
(c) scramjet or
(d) combination of (a), (b), and (c) or
(e) other

Environment Options: 50,000 ft. < h < 70,000 ft.; h = cruise altitude

Design Flight Mach Number: $4 < M < 6$

Compute and Justify: (a) lift
(b) drag
(c) range
(d) endurance
(e) surface heat transfer limits
(f) surface shear stress limits
(g) shock wave boundary layer interaction
(h) material selection
(i) empty weight
(j) gross weight
(k) sketch your configuration
(l) propulsion characteristics

Reference: (a) Hirschel and Welland, 2009, ISBN 978-3-540-89973-0
(b) Bertin, 1994, ISBN 1-56347-036-5
(c) NACA Report 1135, 1953

Hints/Maybe

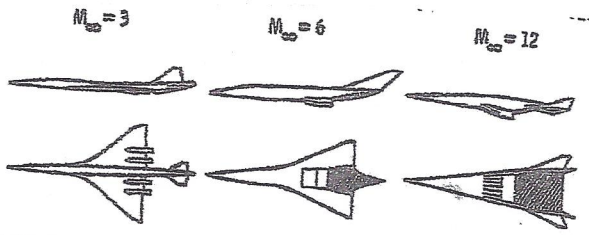

Fig. 4.25. Growth of the size of the external part of the propulsion system (forebody, inlet) and the external nozzle area (hatched) of airbreathing flight vehicles with cruise Mach number M_∞

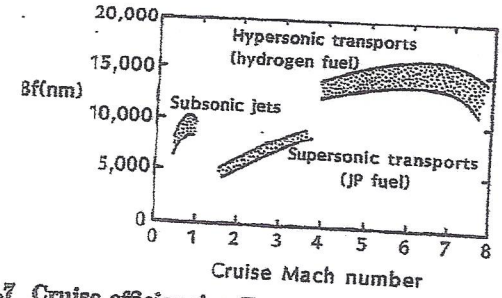

Fig. 10.7 Cruise efficiencies (Breguet factor) of subsonic, supersonic, and hypersonic transports, as taken from Ref. 8.

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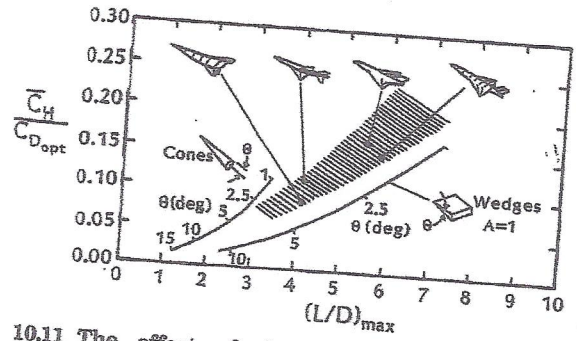

Fig. 10.11 The effect of shape and $(L/D)_{max}$ on heat-transfer/drag ratio evaluations. $M_I = 6$; $Re_L = 140 \times 10^6$; turbulent boundary layer, as taken from Ref. 15.

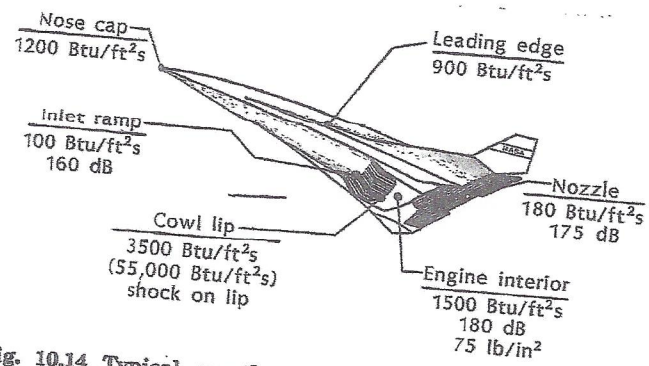

Fig. 10.14 Typical aerothermodynamic constraints for the NASP during ascent, as taken from Ref. 10.

Airbus Patented a Jet Capable of Flying 4 Times the Speed of Sound

- Jack Linshi @jacklinshi

Aug. 3, 2015

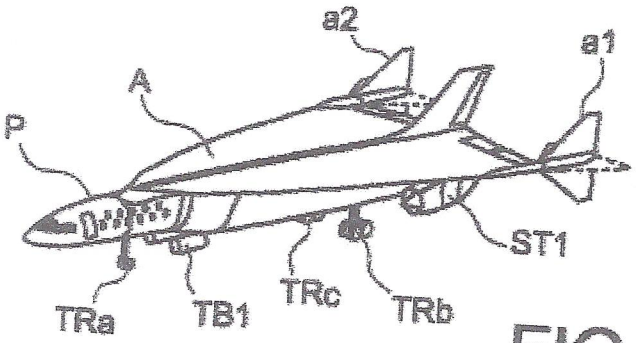

FIG. 15

Airbus/USPTO A perspective view of an ultra-rapid air vehicle according to the invention.

Hyperloop, schyperloop

Airbus has won a patent for an "ultra-rapid air vehicle" that the aircraft maker says could travel over four times the speed of sound, according to the U.S. Patent and Trademark Office.

The patent, which was approved last month, details a hypersonic jet twice as fast as the Concorde, a supersonic jet that had previously been in commercial service. The Concorde, which was developed partly by a company now owned by Airbus, was capable of flying at about 1,300 mph (2,100 km/h), or twice the speed of sound — a feat the new invention can beat with new turbojets and a hydrogen power system, according to the patent.

Like other lofty patents, it's unlikely that the hypersonic jet will become a reality any time soon — at least not within the one-year term for which the patent was approved. Still, some of the ideas involved with the design could make their way to real aircraft further in the future.

http://time.com/3982901/airbus-patent-supersonic-jet/